A Teacher-Led Human-Machine Collaborative Precision Teaching Model With Multimodal Agents for Financial Vocational Education

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Abstract. Financial vocational education faces a persistent tension: classrooms are expected to produce graduates who can interpret business data, judge financial risk, and use intelligent tools, yet students enter with widely different accounting and programming backgrounds. This paper reports the design and pilot implementation of a teacher-led, student-centered precision teaching model supported by multimodal agents in a financial data analytics course. The model organizes teaching into three connected stages: pre-class precise diagnosis, in-class precise instruction, and post-class precise improvement. A group of pedagogical agents collects multimodal learning evidence, builds student profiles, generates differentiated tasks, provides practice support, and produces process-oriented feedback, while teachers retain instructional decisions and final evaluation. The implementation uses a real project, the development of an enterprise financial risk warning agent, as the course carrier. The paper describes the design principles, course transformation, agent roles, collaboration boundaries, and evaluation framework. Preliminary classroom observations suggest that the model is workable in ordinary vocational classrooms and that its value depends on clear task boundaries, teacher control, data compliance, and continuous revision of course resources.

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Applied computing → Education

Applied computing → Computer-assisted instruction

Computing methodologies → Artificial intelligence → Natural language processing

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1. Introduction

The nature of accounting work has changed in the past decade. Financial shared service centers, intelligent risk control systems, and data asset management have moved routine tasks into digital platforms, and the remaining work increasingly requires people who can connect business data with accounting judgment. Chinese vocational colleges are expected to respond to this change, but their teaching practices have not always kept pace. The Vocational Education Branch of the China Commercial Accounting Society placed the design of human-machine collaborative precision teaching models on its 2026 research agenda, together with topics on accounting agents, AI-supported practical training, and accounting industry-education integration [1]. The Education Strengthening Plan (2024-2035) also calls for more intelligent and personalized education [2]. These policy signals make the problem both timely and practical.

The difficulties are visible in ordinary classrooms. Students enter a financial data analytics course with uneven accounting backgrounds, programming experience, and confidence. Some can read a balance sheet but struggle with Python, while others can process data but cannot explain what a financial ratio means. A course that moves at one speed leaves part of the class behind and part insufficiently challenged. Real financial data cannot be distributed freely because of confidentiality, so practical work often relies on simplified cases that do not resemble workplace tasks. Feedback arrives after assignments are marked, and teachers have limited time to diagnose individual mistakes.

Institutional conditions make the problem harder. A typical vocational college course has limited class hours, one teacher, and a heterogeneous cohort. The teacher is also responsible for practical training, competition guidance, and enterprise liaison. Any solution that increases preparation or marking time will not survive contact with the timetable. The model therefore treats teacher workload as a design constraint, not an afterthought.

Large language models and agent-based systems offer a way to make this process more adaptive. A single chatbot can answer questions and generate materials, but it does not by itself produce a coherent teaching cycle. The model proposed in this paper treats multimodal agents as a teaching infrastructure rather than a novelty. The agents collect learning evidence from texts, tables, platform logs, and classroom operations; they build student profiles; they suggest resources and tasks; and they provide process records that teachers can review. Teachers remain responsible for goals, value judgments, and final assessment.

The choice of a financial data analytics course as the test bed is deliberate. The course requires both technical skills and accounting judgment, which makes it a difficult case for personalization. If a precision teaching model can work in this setting, it is likely to transfer to other accounting courses. The project team also reflects this complexity. It includes teachers from big data application, accounting information management, artificial intelligence, and educational technology, because course redesign cannot be done by a single teacher.

This paper makes three contributions. First, it presents a teacher-led, student-centered, agent-supported precision teaching model organized into pre-class diagnosis, in-class instruction, and post-class improvement. Second, it translates the model into a concrete course transformation, using an enterprise financial risk warning agent as the project carrier. Third, it provides an evaluation framework with indicators for knowledge, engagement, project quality, digital literacy, and teacher workload. The empirical work is at an early stage, so the paper reports the design and preliminary classroom observations rather than claiming a completed controlled experiment.

The paper also reports the institutional conditions under which the model can be used, including network access, teacher training, and a maintained knowledge base. These conditions are often ignored in technical papers, but they determine whether an innovation remains in the pilot course or moves into the curriculum.

Three research questions guided the design. How should a multimodal agent group be organized so that it supports diagnosis, instruction, and improvement without taking over the teacher's role? How should a financial data analytics course be restructured so that personalization is connected to a real project? And how can the model be evaluated with evidence that teachers can actually collect? The paper answers these questions through design, implementation, and an evaluation framework, rather than through an abstract discussion of AI in education.

A model is only useful if it can be run by an ordinary teacher. We therefore keep the number of agents small and the interaction simple. The aim is not to build a complete product but to make a teaching process more precise. Each function in the model is tested against a simple question: does it help the teacher see a student's need, does it help the student complete a task, and can it be maintained by a course team without full-time programmers?

A key concern is that the model may be used to justify less teacher attention. We address this concern explicitly. The model does not reduce the time a teacher spends with students; it

changes where that time is spent. Repetitive tasks such as checking completion and grouping errors are handled by the system, while classroom time is used for explanation, discussion, and individual support. The evaluation framework therefore includes teacher workload and contact time as indicators, not only student outcomes.

2. Related Work

2.1. Large Language Models in Education

Research on large language models in education has moved from feasibility demonstrations to more careful accounts of what AI can and cannot do. Rahman and Watanobe examined ChatGPT as a tool for answering questions and assisting engineering teaching [3]. Jeon and Lee described the complementary relationship between human teachers and ChatGPT, arguing that the teacher remains responsible for judgment and emotional support [4]. Usher compared generative AI, instructor, and peer assessment in higher education and found that the form of feedback matters as much as the source [5]. Rus and D'Mello reviewed conversational intelligent tutoring systems and provided an early basis for using dialogue agents in learning [6]. These studies support the idea that AI can reduce routine work, but they do not provide a complete classroom model for vocational education.

From a methodological point of view, evaluations of AI tutoring systems need to separate the tool from the teaching process. A tool that improves test scores in one course may fail in another because the tasks, data, and teacher routines are different. This is especially true in accounting education, where the object of learning is not only a technique but a professional interpretation of financial information.

Several international studies also warn against uncritical use of generative AI. Automated grading and feedback can save time, but the quality of feedback depends on the design of prompts and rubrics. If the underlying model produces vague advice, students may ignore it or copy it without understanding. Our model therefore treats agent feedback as draft material that teachers review. The pedagogical value is not in the model itself but in how the course uses it.

2.2. Human-Machine Collaborative Teaching

In China, studies on AI large models and teachers have produced frameworks for pre-class, in-class, and post-class collaboration. Lin, Liu, and Zhao proposed a human-machine

collaborative teaching strategy and emphasized the role of AI in personalized learning [7]. Feng reviewed task design in collaborative learning and noted that research often underestimates the active role of AI and pays too little attention to subject differences [8]. Xiao, Xiao, and Wang argued that human-machine collaboration changes the teacher's role into that of a learning environment designer and value leader [9]. Tan and Guo reviewed the state, challenges, and trends of AI in vocational education [10]. The general direction is clear, but the operational details, especially the boundary between teacher and agent decisions, remain underspecified.

Chinese research has also emphasized that AI should not weaken the teacher's leading role. The teacher remains responsible for curriculum values, emotional support, and judgment about ambiguous student performance. Our model operationalizes this principle by giving teachers the right to override agent suggestions and by making the final evaluation a teacher decision.

A common limitation in this literature is that AI is described as a single tool. The teacher is told to use AI, but the article rarely explains which functions should be automated, which should remain with the teacher, and how the outputs should be checked. The present paper responds to this limitation by specifying a group of agents with different roles and by making teacher review a required step.

2.3. Precision Teaching and Vocational Curricula

Precision teaching in vocational education usually means using data to diagnose learners, select content, adjust instruction, and evaluate outcomes. This idea aligns with the post, course, competition, certificate integration used in Chinese vocational curricula [11]. In financial education, Zou, Liu, and Liu argued that big data plus collaborative education should combine cross-major, cross-school, and cross-region resources [12]. Gong and Li discussed school-enterprise cooperation in financial education and described joint training bases, shared faculty, and shared training plans [13]. Qi examined the internal mechanism of AI-supported course teaching in vocational colleges [14]. Zhou discussed how generative AI is changing vocational teaching from an auxiliary tool to a component of the teaching system [15]. These studies provide concepts and policy framing, but few of them are anchored in one course with a specific financial project and a measurable teaching loop.

2.4. Research Gap

Three gaps emerge from the literature. First, most studies stay at the level of general education or general vocational education and do not deal with the constraints of financial data and accounting tasks. Second, AI is often treated as a single chatbot or content generator, not as a set of agents with distinct functions in a teaching process. Third, precision teaching is described as an ideal but is rarely connected to course modules, classroom activities, and assessment instruments. This paper addresses these gaps by specifying agent roles, a three-stage loop, a financial course redesign, and an evaluation framework.

From a technical perspective, the agent group can be built with a low-code platform, a vector database, and a set of prompt templates. The technical choices are deliberately kept simple because the target users are teachers and vocational students, not software engineers. A complex architecture would make the model harder to maintain and would distract from the teaching problem.

The accounting context adds an important constraint that generic education research often ignores. Financial data are sensitive, accounting standards change, and professional judgment cannot be reduced to a single probability. A student may calculate an indicator correctly and still misread what it means for a particular firm. The model therefore separates computational support from interpretive support. Agents can check formulas and retrieve standards, but the interpretation of a risk signal remains a classroom discussion led by the teacher.

3. Design of the Model

3.1. Design Principles

The model follows six design principles. The teacher is the instructional decision maker. The student is the learner and remains responsible for producing work. Agents support diagnosis, resource selection, practice, and feedback, but they do not replace teacher judgment. The teaching process must form a closed loop from diagnosis to intervention to evaluation. The tools must reduce teacher workload rather than add a parallel administrative burden. Data use must be transparent, consent-based, and appropriate for the financial education context.

The knowledge base is not static. It stores financial ratios, accounting standards, case data, common errors, and prompt templates. Each semester, teachers add new cases and revise rules after reviewing agent performance. This maintenance is a normal part of the model, not an additional project. A college with limited resources can begin with a low-code platform and a

small number of agents. The model does not depend on one commercial product, which is important for transfer.

The framework can be implemented at different levels of technical intensity. A minimal version uses a spreadsheet of learning records, a small set of prompt templates, and teacher review forms. A more advanced version uses a low-code platform, a knowledge base, and dashboard visualizations. Both versions preserve the same pedagogy, which means the model can be introduced gradually.

3.2. Overall Framework

The overall framework has three actors and three stages. The actors are the teacher, the student, and a group of multimodal agents. The stages are pre-class precise diagnosis, in-class precise instruction, and post-class precise improvement. The three stages run on a common line of work, in this paper the development of an enterprise financial risk warning agent. The teacher sees profiles and process records in a dashboard rather than reading every conversation. The agents draw on a knowledge base that contains financial standards, course materials, enterprise cases, and common student errors. Figure 1 shows the framework.

The three stages are not isolated. Each stage produces an output used by the next: profiles feed task design, classroom records feed homework, and homework results update the profile. The loop can run weekly or per module. This continuous cycle is what distinguishes precision teaching from occasional personalization.

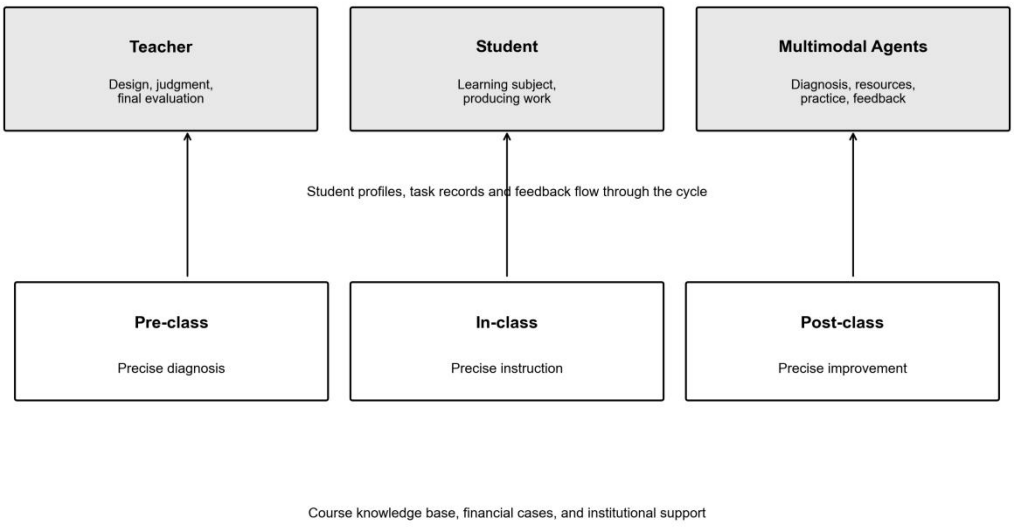


Figure 1. Overall framework of the teacher-led human-machine collaborative precision teaching model.

3.3. Roles of the Multimodal Agents

The agent group is not a single chatbot. Different agents are responsible for data collection, resource recommendation, practice support, error diagnosis, and evaluation reporting. In the pre-class stage, agents collect prior grades, pre-test results, homework, and platform operations. In the in-class stage, they generate cases, support layered tasks, and answer routine questions. In the post-class stage, they create personalized homework, classify errors, and summarize progress. Table 1 links these functions to teacher responsibilities.

The design of agent roles is guided by the workload of a real teacher. Tasks that are repetitive, such as collecting records and grouping errors, are given to agents. Tasks that require professional judgment, such as interpreting a student's unusual performance or deciding how to handle a sensitive financial case, remain with the teacher. This division is more useful than a generic statement that AI should assist teaching.

Stage	Core task	Agent functions	Teacher responsibilities
Pre-class	Profile and grouping	Collect tests, homework and logs; build a profile; suggest levels	Review profiles; set goals and groups
In-class	Differentiated practice	Generate cases; give code and format hints; record progress	Explain concepts; manage tasks; guide risk judgment
Post-class	Diagnosis and extension	Group errors; recommend tasks; form an evaluation report	Check the report; arrange support; adjust later teaching

Table 1. Agent roles and teacher responsibilities in the three teaching stages.

3.4. Pre-Class Precise Diagnosis

The pre-class stage begins with data collection, not with a lecture. The agent system gathers prior course results, pre-test answers, preview records, and submitted homework. The profile is updated at the start of each module, not once per term. If a student repeatedly changes a solvency ratio formula, the system marks that knowledge point as weak; if the same type of error appears across modules, the profile recommends individual attention. The teacher reviews the profiles and groups students into foundation, application, and extension levels.

Profiles are presented as evidence, not labels. A student is not classified as weak or strong; the profile lists specific gaps and recent progress. Teachers use the profile to decide which tasks to assign, but they can override the suggestion. The diagnosis is therefore a starting point for professional judgment, not an automated verdict.

The profile also includes a professional quality dimension. For financial courses, this means observing whether students follow data rules, respect confidentiality, and document their assumptions. These behaviors are difficult to measure with a single test, but they are visible in platform logs and student work. Including them in the profile makes the precision teaching model more relevant to accounting education.

The student profile is not intended to predict grades. It is intended to make teaching decisions visible and to help teachers compare what they believed about a student with what the records show. A teacher may notice, for example, that a student who performs poorly on tests completes every practical task carefully. That observation can change the support strategy more than a single average score.

3.5. In-Class Precise Instruction

During the class, the teacher opens with a simplified enterprise case and explains the business objective of financial risk warning. Agents push different data samples and task cards to groups based on the profiles. Foundation students complete data cleaning and indicator calculation, application groups analyze risk and causes, and extension groups adjust model parameters and write a risk report. Agents answer routine questions, correct code, and remind students about report structure.

The teacher concentrates on concepts, risk discussion, and group progress. Agents do not tell students whether a financial judgment is simply right or wrong; they provide evidence and prompts, and the teacher directs the interpretation. The task boundaries are flexible. Foundation students can move to comprehensive tasks after completing the required steps, and comprehensive groups can return to basic tasks to check their work. The teacher watches for two problems: students copying agent suggestions and students becoming dependent on the tool. These problems are detected through classroom observation and work review.

3.6. Post-Class Precise Improvement

The post-class stage connects homework, review, and extension. Agents generate personalized assignments from classroom records. Errors are grouped by knowledge point,

and the system recommends short videos, formula cards, or practice tasks. Weaker students receive compensatory exercises, while stronger students continue with the enterprise risk report or an agent optimization task. The teacher checks the automatically generated evaluation report, pays attention to high-risk students and ambiguous cases, and decides whether to arrange individual support.

The report uses evidence and avoids labeling students with a single grade. It contains knowledge mastery, task completion, collaboration, and progress, with concrete examples behind each suggestion. The teacher checks whether the agent missed an important performance and whether missing data caused an inaccurate judgment. The results of this check are used to improve the agent rules.

3.7. Collaboration Boundaries

The boundaries are as important as the functions. Agents do not assign final grades, do not make value judgments about students, and do not treat model output as the only source of truth. Personal data are collected only after students are informed and agree. Financial data are used in de-identified teaching cases. The knowledge base is updated when accounting standards or financial policies change. Teachers remain responsible for final evaluation, and their classroom observations and student work are part of the evidence.

Ethics and data compliance are part of the model. Students must know what data are collected and how they will be used. Financial cases must be de-identified before they enter the knowledge base. Agent outputs should be traceable so that teachers can see which prompt and source material produced a suggestion. These rules are written into the course implementation manual, not left to individual teachers to improvise.

The agent system should be tested with small pilot groups before full use. The pilot period gives the course team time to check whether the prompts produce useful output, whether the dashboard is readable, and whether the system adds work instead of removing it. Problems found in a pilot are easier to fix because the number of students is small and the course team can compare records with their own observations.

4. Implementation in a Financial Data Analytics Course

4.1. Course and Project Carrier

The model was implemented in a big data financial analysis course for a vocational college in China. The course was redesigned around a single project: developing an enterprise financial risk warning multimodal agent. Students played two roles at the same time. They acted as financial analysts who interpreted indicators, and as designers who specified what the agent should do, what data it should use, and how it should explain a warning.

The project gives the course a natural progression. In the first modules, students learn to prepare data and compute indicators. In later modules, they use the indicators to build a warning model and then design the agent interface. The final report asks them to explain their choices to a manager. This progression turns the course into a portfolio rather than a set of disconnected exercises.

The course team rewrote the syllabus around outcomes rather than chapters. Each outcome is expressed as a task that produces evidence: clean a dataset, calculate indicators, explain a warning, build an agent, write a report. The textbook is used as a reference, but the class is organized by the project. This change is visible to students, who can see why each task matters.

An example task can illustrate the design. In the financial risk identification module, students receive a small enterprise dataset and an industry benchmark. Foundation students calculate three ratios and describe whether each one signals risk. Comprehensive students compare the ratios with the benchmark and identify possible causes. Extension students propose two features for a warning model and explain why those features might be useful. The same dataset supports three levels of work, and the agent system can suggest which level a student should start with.

4.2. Course Modules

Table 2 lists the six course modules. Each module has foundation, comprehensive, and extension tasks. Foundation tasks are required for all students. Comprehensive tasks require judgment on an enterprise case. Extension tasks allow students to choose a financial, technical, or product direction. This structure supports the profiles created in the pre-class stage.

Table 2. Course modules and intended abilities.

Module	Main content	Intended ability
Financial data collection and cleaning	Reports, transaction data, missing values	Data quality and processing
Financial indicator calculation	Solvency, profitability, operation, growth	Financial business understanding
Financial risk identification	Indicator changes, industry comparison, clues	Analysis and judgment
Risk warning model construction	Features, training, interpretation	Modeling and technical application
Agent function design	Prompts, knowledge base, interaction	Agent design and collaboration
Risk report and review	Structure, visualization, case review	Communication and transfer

4.3. Teaching Organization

The teaching process follows a pre-class task, in-class project, post-class iteration pattern. Before class, students complete a platform task and the agent system updates the profile. During class, small groups work on the enterprise risk warning task with support from the teacher and agents. After class, students revise their models and reports based on feedback, and strong work enters a case library. Most practical sessions are held in a computer room so that students can move between data operations, model debugging, and agent interaction in one environment.

Each module has a small assessment point. The teacher can see how many students complete each foundation task and where common errors occur. This makes it possible to adjust the next class session rather than waiting until the end of the term. The same data are also used to update the next pre-class profile, which is why the loop remains closed.

Group composition changes over the term. The teacher uses the latest profile to adjust groups, so a student who improves in one module is not locked into the same level for the whole course. This dynamic grouping is one of the practical benefits of keeping the profile current.

The course also includes a final showcase. Each group presents its risk warning agent, explains the data and indicators it uses, and describes one limitation. The showcase serves as a summative assessment and as a way to connect classroom work with enterprise expectations. It also gives students practice in explaining technical choices to non-specialists, which is a common requirement in financial roles.

Students also need training in using agents responsibly. The course begins with a short module on what agents can and cannot do. Students learn to check references, compare agent suggestions with course materials, and record the assumptions behind their final answers. This

training is not an extra topic; it is part of the professional behavior expected in a financial workplace.

4.4. Assessment

Assessment combines process data and ability outputs. Process data include preview completion, task completion, platform interaction, error types, and revision counts. Ability outputs include the risk analysis report, the agent work, data visualizations, and a presentation. Teacher evaluation, student self-assessment, and agent process records contribute to the final grade. Table 3 lists the evaluation dimensions.

Dimension	Indicators	Data source
Knowledge mastery	Pretest, posttest, unit tests	Platform scores
Learning engagement	Preview rate, interaction, completion	Platform logs
Ability output	Risk report, agent work, visualizations	Teacher assessment
Professional quality	Data compliance, collaboration, presentation	Observation and peer review

Table 3. Evaluation dimensions, indicators, and data sources.

Course teams should decide the weights at the beginning of the term. The important point is that students do not rely on a single final examination and that process behavior is visible. Each module result feeds into the next module, so the assessment is part of learning rather than an end-of-term event.

The evaluation report is shared with students before the final grade is assigned. Students can see which evidence was used, challenge missing records, and submit additional work. This transparency reduces the feeling that the system is grading them secretly. It also encourages students to review their own process, which is a skill they will need in professional life.

4.5. Industry-School Integration and Credentials

The course content is aligned with financial data analysis job requirements and with certification content used in Chinese financial digitalization. Enterprises review cases, join project evaluation, and provide internship observation. Competition tasks can use the risk warning report and the agent work produced in class. Industry-school cooperation is useful when it changes the course, not only when it adds a signing ceremony.

The cooperation process has four steps. Enterprises provide de-identified cases and job standards; teachers and enterprise experts break the cases into tasks; students receive project review and workplace observation; outstanding work enters a course case library. In this way, the course resources are updated continuously rather than remaining static.

5. Evaluation Design and Preliminary Results

5.1. Study Design

The evaluation plan compares an experimental class using the model with a control class using the original teaching approach. The two classes should be similar in prior achievement, use the same teacher, course hours, content, and assessment standards, and differ only in teaching organization and the use of agents. The study collects pretests, posttests, platform logs, classroom observations, student questionnaires, and teacher interviews.

To make the comparison fair, the course team must ensure that the control class also receives the same content and opportunities for feedback. The experimental class is not given extra hours or better materials. The only intended difference is the teaching model. This condition is important for the credibility of any later quantitative comparison.

The sample size and class composition will depend on the institution's enrollment. In a small vocational course, a randomized design may not be possible. In that case, the course team can use two parallel classes taught in the same term, or a sequential comparison across terms. The evaluation framework should be adapted without lowering the standard of evidence.

5.2. Indicators

The indicators cover knowledge mastery, learning engagement, project quality, digital literacy, and teacher workload. Knowledge mastery comes from pretest and posttest scores. Engagement comes from preview rates, interaction frequency, and task completion. Project quality comes from the risk report and agent work. Digital literacy is observed through data handling, visualization, and tool use. Teacher workload is recorded through preparation and marking time and through interviews.

The questionnaire includes items on perceived difficulty, usefulness of agent hints, willingness to continue using the tool, and confidence in explaining financial data. These subjective measures are combined with platform logs and work scores. Teacher interviews

focus on how much time the agent system saves, which agent suggestions are useful, whether students depend too much on the system, and what institutional support is needed.

The study will also collect a small number of open-ended student comments. These comments help the research team understand why a student found a task too difficult or why an agent suggestion was not useful. Open-ended comments are coded together with the teacher interviews, and the results are used to revise prompt templates and task cards.

The control condition should be described in enough detail for readers to judge whether the comparison is fair. If the control class uses a different textbook or receives additional help, the comparison loses its meaning. The course team therefore records the teaching plan for both classes and publishes the materials in an appendix. This transparency also helps other schools decide whether the model can work in their context.

5.3. Analytical Strategy

The analysis proceeds at three levels. Descriptive statistics compare averages, completion rates, and satisfaction distributions. Difference tests examine whether changes between pretest and posttest are statistically meaningful. Case analysis follows students who improve sharply or show recurring problems, using classroom records to explain the pattern. The three levels are intended to prevent conclusions based on a single score.

The evaluation framework is designed to be reusable. Other schools can keep the same dimensions and change the weights according to their curriculum. The table can be turned into a dashboard that shows the teacher which indicators need attention. The framework does not prescribe one statistical test, because sample sizes and course conditions will differ across institutions.

The platform logs need a data dictionary before the course begins. Each variable should have a clear name, unit, collection time, and meaning. Without this preparation, teachers may find themselves trying to interpret ambiguous records at the end of the term. The data dictionary is part of the course implementation, not a research afterthought.

5.4. Preliminary Observations

Preliminary implementation supports the feasibility of the model, although the controlled data collection has not been completed. The agent system makes it easier for teachers to see common problems because platform logs concentrate evidence that was previously scattered

across homework and tests. Layered tasks give students a clearer next step, and the project structure gives the course a visible product.

Classroom observation also shows two risks. Some students copy agent suggestions without understanding them, and some expect the agent to make decisions for them. Teachers therefore need to check work quality, ask students to explain their choices, and treat the agent output as material for discussion rather than as an answer. The model works only when students know that agent suggestions are evidence and not answers.

To make the results transparent, the paper will report the number of students, missing data, and any differences between classes before the intervention. A table will show pretest means, posttest means, and change values, with the raw data available in an appendix. This reporting standard is necessary because small convenience samples are common in vocational education research.

5.5. Limitations

The current evidence has clear limits. It comes from one course in one institution, and parts of the data table have not been completed. The evaluation focuses on short-term outcomes and cannot yet show long-term effects on employment or professional development. The system depends on network access, accounts, and institutional support. Some vocational colleges have weak information infrastructure, and the cost of keeping models and knowledge bases current needs further study.

6. Discussion

6.1. Contributions

The main contribution is a concrete way to organize human-machine collaboration in financial vocational education. The model makes the boundary visible: agents collect and process, teachers decide and evaluate, students produce and explain. This boundary is often missing in studies that simply recommend AI-assisted teaching. The course design also shows how a real financial project can carry a teaching loop rather than remain a separate exercise.

The three-stage loop is not a sequence of isolated tools. Each stage produces an output used by the next: profiles feed task design, classroom records feed homework, and homework results update the profile. The loop can run weekly or per module. This continuous cycle is what distinguishes precision teaching from occasional personalization.

The model also offers a way to talk about educational equity. Students with weaker backgrounds receive more targeted support through profile updates and compensatory tasks. Students with stronger backgrounds receive extension work instead of waiting for the class to catch up. The system does not eliminate differences, but it gives the teacher a practical way to respond to them.

6.2. Comparison With Prior Work

Compared with prior work, this paper is more specific about the division of labor. Rahman and Watanobe and Jeon and Lee discuss the value of large language models, but they do not specify how a set of agents should operate across a course [3,4]. Lin et al. propose a pre-in-post framework, but they do not connect it to financial data, risk judgment, and certification requirements [7]. The framework in this paper borrows from these studies and adds the financial vocational context.

One lesson from the pilot is that the quality of prompt templates and the knowledge base matters more than the choice of the underlying large language model. If the agent returns vague advice, students ignore it or copy it. The course team therefore writes prompts with a specific financial task, constraints, and output format. The same principle applies to feedback: a clear rubric is more useful than a generic explanation.

The paper does not claim that agents are necessary for every teaching action. In many cases, a well-designed worksheet and a careful teacher can achieve the same result. The value of the agent system is that it scales the collection and organization of evidence, freeing the teacher to spend time on interpretation and personal interaction. The decision to use agents should be based on this trade-off, not on fashion.

6.3. Conditions for Transfer

Transfer depends on three conditions. Teachers must be willing to redesign instruction and to read profiles and process records, not simply to add a tool. Institutions must provide stable network access, accounts, and technical support. The course team must maintain the case library and knowledge base. The model itself does not require a specific commercial platform, so schools can adapt the implementation manuals and evaluation instruments to their local conditions.

The findings also have policy implications. Teacher training should move from showing how to use a chatbot to practicing how to interpret student profiles and review agent reports.

Institutional quality systems should accept process data as part of course evaluation. Financial data compliance should be treated as a curriculum topic rather than an obstacle, because it is exactly the kind of professional judgment that students need.

Teacher development should therefore be part of the implementation plan. Training sessions should include how to read a profile, how to review an agent report, and how to design a task that gives the agent useful data. Without this training, the same tool can produce different results in different classrooms. The model is not a substitute for teacher preparation; it depends on it.

7. Conclusion

This paper described a teacher-led, student-centered precision teaching model supported by multimodal agents and its application to a big data financial analysis course. The model links pre-class diagnosis, in-class instruction, and post-class improvement into a closed loop, uses an enterprise financial risk warning project as the course carrier, and defines the roles of teachers, students, and agents.

The implementation includes six course modules, layered tasks, process-based assessment, and an evaluation framework covering knowledge, engagement, project quality, digital literacy, and teacher workload. Preliminary observations suggest that the model is feasible in ordinary vocational classrooms and that its success depends on clear task boundaries and teacher control.

Future work should complete the controlled comparison, extend the model to other accounting courses, examine long-term student outcomes, and explore lightweight deployment for schools with limited information infrastructure. The long-term goal is to turn the model into reusable course resources, implementation manuals, and evaluation tools for financial vocational education.

Because the model is designed for transfer, the course team documents not only what worked but also what had to be adjusted. This documentation includes changes to task cards, prompt templates, group sizes, and assessment weights. The records make the model easier to improve and easier for another institution to adapt.

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